ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE C

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Resistors and the bridge rectifier; DC and servo motors and PLC ladder programs; hydraulic filters, cylinder forces and two cylinders working together; electro-hydraulic timed control; and maintenance.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Draw all circuits and symbols neatly, using a ruler and standard symbols.
5. Use only a black or blue pen.
6. Write neatly and legibly.

QUESTION 1

1.1 Define the term *resistor*. (2)

1.2 A resistor has the colour bands brown, black, red and gold. State its value and its tolerance. (3)

1.3 Differentiate between *alternating current (AC)* and *direct current (DC)*. (2 × 2) (4)

1.4 Draw a fully labelled circuit diagram of a full-wave bridge rectifier with a smoothing capacitor and a load resistor. (6)

1.5 A 100 Ω resistor and a 150 Ω resistor are connected in parallel across a 12 V supply. Calculate:

1.5.1 The total resistance (3)

1.5.2 The total current (2)

[20]

QUESTION 2

2.1 Define the term *servo motor*. (2)

2.2 Explain how the direction of rotation of a DC motor is reversed. (2)

2.3 Name THREE main parts of a DC motor. (3)

2.4 FIGURE 1 below shows a PLC ladder program that controls a pump (Q0.0) and a lamp (Q0.1). I0.0 is a START button (NO) and I0.1 is a STOP button wired as normally closed.

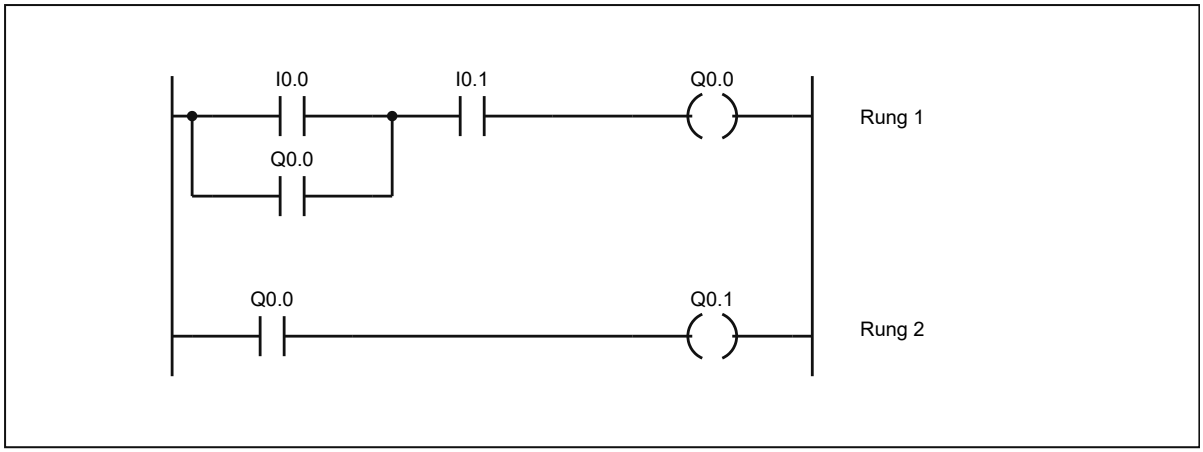


FIGURE 1

2.4.1 What happens to Q0.0 and Q0.1 when I0.0 is pressed and released? (2)

2.4.2 Explain the function of the Q0.0 contact in rung 1. (2)

2.4.3 Explain what happens when I0.1 is pressed. (2)

2.5 Draw a PLC ladder diagram for a pump (Q0.0) that is started by I0.0 (NO) and stopped by I0.1 (NC). A float switch I0.2 closes when the tank is full and must also stop the pump. A lamp (Q0.1) must show that the tank is full. (7)

[20]

QUESTION 3

3.1 Explain the function of a filter in a hydraulic system and name TWO places where it can be fitted. (3)

3.2 A double-acting cylinder has a bore of 80 mm and a piston-rod diameter of 40 mm. It works at a pressure of 10 MPa. Calculate:

3.2.1 The force when the cylinder extends (3)

3.2.2 The force when the cylinder retracts (3)

3.3 Draw a hydraulic circuit in which two double-acting cylinders, A and B, extend and retract together, controlled by ONE manually operated 4/3-way valve with a tandem centre. Include a hydraulic power pack and a pressure gauge. (11)

[20]

QUESTION 4

4.1 Explain the purpose of a limit switch in an electro-hydraulic system. (2)

4.2 Differentiate between a *direct-operated* and a *pilot-operated* solenoid valve. (2 × 2) (4)

4.3 A press cylinder is controlled by a 4/2-way valve with a single solenoid (Y1) and a spring return. Design the electro-hydraulic working circuit and the electrical control circuit so that: pressing S1 makes the cylinder extend (relay K1 holds itself); when the cylinder is fully extended, limit switch S2 starts an on-delay timer T1; after 3 seconds T1 releases K1 and the cylinder retracts; and an emergency stop removes the power from the control circuit. (14)

[20]

QUESTION 5

5.1 Explain the term *predictive (condition-based) maintenance*. (2)

5.2 Explain how you would check a PNP proximity sensor with a multimeter. (4)

5.3 State FIVE causes of overheating of the oil in a hydraulic system. (5)

5.4 Explain, in the correct order, the lock-out and tag-out procedure before working on a machine. (5)

5.5 Give FOUR reasons why maintenance records must be kept. (4)

[20]

TOTAL: 100

END OF QUESTION PAPER